SEVEN FIELDS · 56 EXPLICIT CURVE MODELS · EXACT INVARIANTS

# Small finite fields and elliptic-curve structure

## Isogenies, discriminants, endomorphisms, twists, and embedding degrees

**The curves can be grouped, but several different groupings are involved.** Over one finite field, matching point counts identifies an isogeny class. Matching $j$ identifies geometric isomorphism, often with different twists. Matching an endomorphism order is another relation. A sparse modulus alone does not identify any of these groups.

The report compares the four prime fields discussed in the conversation and the three binary fields with degrees 53, 79, and 83. No curve equations were supplied, so the examples include deliberately structured CM/Koblitz models, general-coefficient controls, and two screened 83-bit examples. "Cisogenes" and "inventing degree" are interpreted as *isogenies* and *embedding degree*.

**The main findings**

- Over $p = 2^{83} - 2^{59} - 1$, all four special-$j$ models in the sample are supersingular, have the same order $p+1$, and are isogenous over $\mathbb{F}_p$. Their largest prime factor is only 241. This follows from congruences and the smoothness of $p+1$.
- The binary Koblitz models have endomorphism discriminant $-7$ in all three fields. Their much larger Frobenius discriminants come from conductor squares. This apparent clustering is explained by descent from $\mathbb{F}_2$.
- A useful prime-field example is $y^2 = x^3 - 3x + 154$ over $2^{83} - 55$. It has an 83-bit prime order, a large embedding degree, a large fundamental CM discriminant, and an 81-bit prime subgroup on its quadratic twist.
- General-coefficient controls have different, large CM discriminants, yet several have badly factored group orders. Large discriminant, large embedding degree, and large prime subgroup are separate requirements.

**What this establishes.** The measured clustering is accounted for by standard isogeny, CM, twist, and Frobenius structure. The sample does not enumerate all curves or prove that every curve over a selected field behaves similarly. These small fields support experiments; even the best subgroup here has only about 41.5 bits of generic discrete-logarithm work.

01 / SCOPE

# The fields and the exact example models

For a prime field the integer modulus specifies $\mathbb{F}_p$. For a binary field, an irreducible polynomial specifies a convenient presentation of $\mathbb{F}_{2^m}$; it does not create a new abstract field of that size. All four primes were certified by Lucas primality certificates. The binary polynomials passed Rabin irreducibility tests.

| Label | Prime modulus | Decimal modulus | $g$ |
|---|---|---|---|
| `p61` | $2^{61}-1$ | 2305843009213693951 | 37 |
| `p64` | $2^{64}-2^{32}+1$ | 18446744069414584321 | 7 |
| `p83s` | $2^{83}-2^{59}-1$ | 9671405980456281094225919 | 11 |
| `p83` | $2^{83}-55$ | 9671406556917033397649353 | 7 |

Here $g$ is a primitive root, hence a nonsquare, used to name the special-$j$ twist classes. `p83s` labels the three-term Solinas prime; `p83` labels the pseudo-Mersenne prime $2^{83}-55$.

| Label | Cardinality | Irreducible polynomial $f_m(X)$ |
|---|---|---|
| `b53` | $2^{53}$ | $X^{53}+X^6+X^2+X+1$ |
| `b79` | $2^{79}$ | $X^{79}+X^9+1$ |
| `b83` | $2^{83}$ | $X^{83}+X^7+X^4+X^2+1$ |

In each binary field, $u$ is the residue class of $X$ modulo $f_m(X)$.

**Prime-field families**, all in the model $y^2=x^3+Ax+B$:

- E4: $A=-g^i$, $B=0$, with $0\le i<\gcd(4,p-1)$; $j=1728$. These give every $\mathbb{F}_p$-isomorphism class at this $j$.
- E3: $A=0$, $B=g^i$, with $0\le i<\gcd(6,p-1)$; $j=0$. These give every $\mathbb{F}_p$-isomorphism class at this $j$.
- G: $A=-3$, $B=1$, and its quadratic twist by $g$; $j=2304$.
- S: the screened `p83` model $A=-3$, $B=154$, and its quadratic twist by 7.

**Binary families**, all in the model $y^2+xy=x^3+ax^2+b$: K uses $b=1$; B uses $b=u$; the screened S pair uses $b=u^{251}$ in degree 83. Each includes $a=0$ and $a=1$. The label `B83-a1` is a report label for a B-style example, not a standardized NIST curve.

02 / GROUPING

## Which relations really group the curves?

| Grouping | Correct test | What a match means |
|---|---|---|
| Same finite field | Same characteristic and cardinality | Different field presentations can still be isomorphic. |
| Geometric isomorphism | Same $j$ in a common algebraic closure | Includes twists; point counts over the base field can differ. |
| Base-field isogeny | Same $q$ and trace $t$ | Equivalent to the same $\#E(\mathbb{F}_q)$, by Tate's theorem [1]. |
| Ordinary CM field | Same fundamental $D_K$ | Same imaginary quadratic algebra, not necessarily the same order or isogeny class. |
| Ordinary endomorphism order | Same $D_K$ and conductor $f_E$ | Same abstract order; still does not force a common base-field trace. |
| Rational point group | Same invariant factors $d_1 \mid d_2$ | An abelian-group match; not by itself a field/curve/isogeny identification. |

The prime fields here have different characteristics. There is no single common finite-field base over which their curves can be declared isogenous. Models may instead be reductions of the same curve over $\mathbb{Q}$ or share a CM order. The binary fields do share an algebraic closure of $\mathbb{F}_2$, but an isogeny claim must still specify its field of definition.

For a prime-field quadratic twist, a nonsquare $d$ changes $(A, B)$ to $(d^2A, d^3B)$ and reverses $t$. Geometric automorphism groups in characteristic greater than 3 have orders 6 at $j = 0$, 4 at $j = 1728$, and 2 otherwise. The ordinary binary examples have geometric automorphism order 2.

**A concrete isogeny.** For every prime in this report, the curve

$$E : y^2 = x^3 - x \quad \longrightarrow \quad E' : Y^2 = X^3 + 4X$$

has the degree-two isogeny

$$\phi(x, y) = \left( x - \frac{1}{x},\ y\left(1 + \frac{1}{x^2}\right) \right).$$

The map extends at $x = 0$ and infinity; its kernel is $\{O, (0, 0)\}$. Substitution verifies the target equation. The existence of this map forces equal point counts over the base field. For the two primes congruent to 3 modulo 4, the two models are nonisomorphic but isogenous; their point groups need not be isomorphic.

**An especially visible grouping.** Over `p83s`, the two $j = 0$ classes and the two $j = 1728$ classes all have $t = 0$. Thus all four are in one $\mathbb{F}_p$-isogeny class, although their $j$ values and some rational endomorphism orders differ.

**Isogeny graphs.** For an ordinary curve and a prime $\ell$ different from the characteristic, rational $\ell$-isogeny graphs have the familiar volcano structure [3]. Horizontal edges preserve the endomorphism order; vertical edges change its conductor. A single-$\ell$ graph can have several components within one full isogeny class. This report proves class membership via traces and gives the explicit map above; it does not compute every shortest isogeny path.

03 / DISCRIMINANTS

# Four quantities that must be kept separate

For $p > 3$, the curve model has

$$\Delta_E = -16(4A^3 + 27B^2), \qquad j = 1728\frac{4A^3}{4A^3 + 27B^2}.$$

These are field elements, and $\Delta_E \neq 0$ is the nonsingularity test. The curve discriminant changes by a twelfth power when coordinates change. In the ordinary binary model above, $\Delta_E = b$ and $j = b^{-1}$.

Write $N = \#E(\mathbb{F}_q)$ and $t = q+1-N$. Frobenius has characteristic polynomial $T^2 - tT + q$, and its *integer* discriminant is

$$D_\pi = t^2 - 4q.$$

For an ordinary curve, factor this uniquely as

$$D_\pi = f_\pi^2 D_K, \qquad K = \mathbb{Q}(\sqrt{D_K}), \quad \mathbb{Z}[\pi] \subseteq \mathrm{End}(E) \subseteq \mathcal{O}_K.$$

Here $D_K$ is fundamental, and the actual endomorphism order has conductor $f_E \mid f_\pi$ and discriminant $D_{\mathrm{End}} = f_E^2 D_K$. The trace generally identifies the possible orders, not the actual one [2]. When $f_\pi = 1$, maximality is automatic. Additional model/torsion arguments prove maximality for the other ordinary examples used here.

| Control / screened model | $D_K$ | $f_\pi$ | $f_E$ |
|---|---|---|---|
| `p61-G-0` | $-725493073255807$ | 62 | 1 |
| `p64-G-0` | $-12684772546625937963$ | 1 | 1 |
| `p83s-G-0` | $-96266547841332693335441519$ | 2 | 1 |
| `p83-G-0` | $-36997680785363122195732443$ | 1 | 1 |
| `B53-a0` | $-21521076074067967$ | 1 | 1 |
| `B79-a0` | $-24039825893256472878448 63$ | 1 | 1 |
| `B83-a0` | $-89806536697543099455406 07$ | 1 | 1 |
| `p83-S-0` | $-38605735915768438953606187$ | 1 | 1 |
| `S83-a1` | $-38456677135816615942911871$ | 1 | 1 |

The degree-61 G control is instructive: $f_\pi = 62$ but $f_E = 1$. Full rational 2-torsion excludes a factor 2 in $f_E$; an exact test on the degree-480 division polynomial for 31-torsion excludes 31. Its twist has the same ordinary endomorphism order.

**Supersingular distinction.** The geometric endomorphism ring is a maximal order in the quaternion algebra ramified at the characteristic and infinity. Its base-field rational subring can be quadratic. For the prime-field $t = 0$ examples here, the rational ring is either $\mathbb{Z}[\sqrt{-p}]$ or $\mathcal{O}_{\mathbb{Q}(\sqrt{-p})}$, distinguished by whether all of $E[2]$ is $\mathbb{F}_p$-rational. The appendix's supersingular $D_{\mathrm{End}}$ always refers to this rational quadratic ring; it is not the geometric quaternion-ring discriminant. No individual quaternion-order basis is computed.

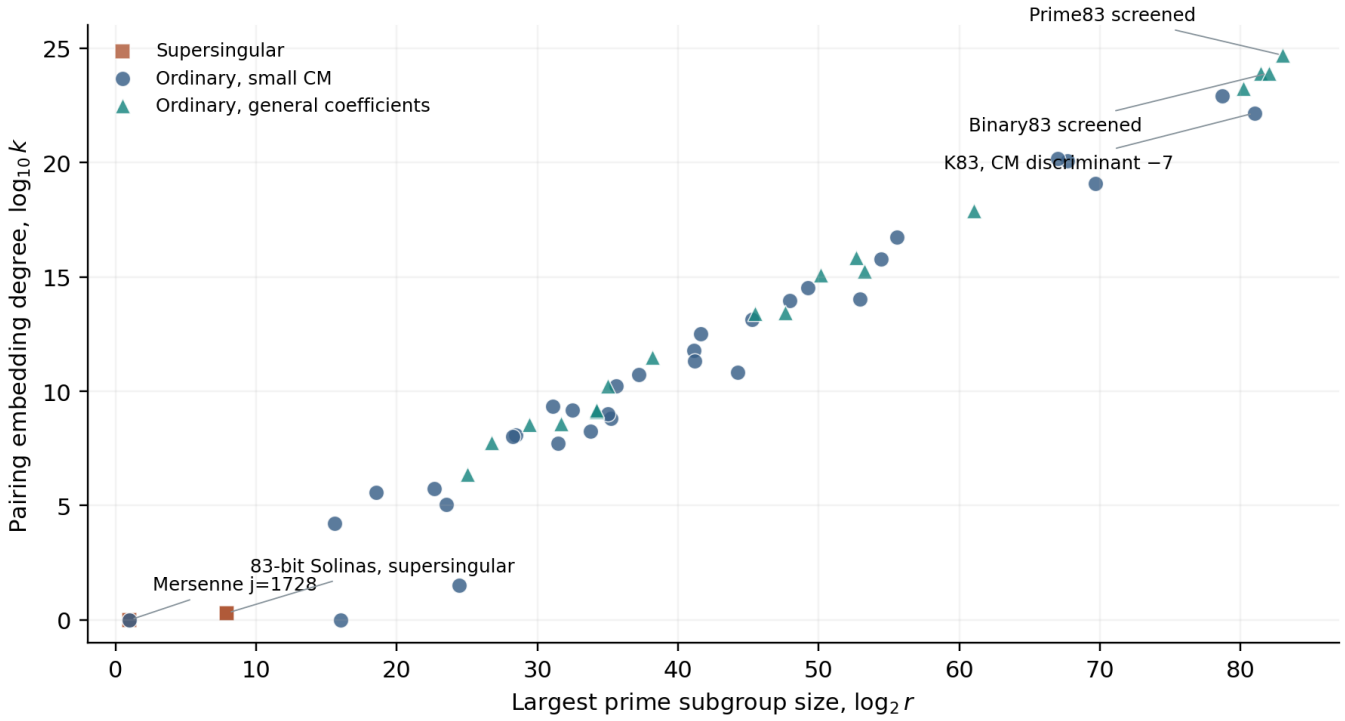
04 / EMBEDDING DEGREE

# Large embedding degree is one independent test

For a specified prime subgroup order $r \mid N$, with $r$ different from the characteristic, the pairing/MOV embedding degree is

$$k = \mathrm{ord}_r(q) = \min\{k > 0 : q^k \equiv 1 \pmod{r}\}.$$

Every reported $k$ uses the *largest prime factor* of $N$. Its exactness is checked by $q^k \equiv 1$ and $q^{k/\ell} \not\equiv 1$ for every prime $\ell \mid k$. Small subgroups on the same curve may have different embedding degrees. The minimum field defining full $r$-torsion can require separate treatment in exceptional cases, notably $k = 1$.




The plot shows all 56 records; coincident points overlap. Its horizontal axis is subgroup size, not field size. Large $k$ does not repair a small $r$: several controls have a very large embedding degree but a weak largest subgroup.

**Three independent measurements:** $r$ determines generic subgroup work; $k$ controls the usual low-degree pairing transfer; $D_K$ and $D_{\mathrm{End}}$ describe CM/endomorphism structure. None can substitute for the other two.

Supersingular prime-field controls have $N = p + 1$. For any odd prime $r \mid N$, $p \equiv -1 \pmod{r}$ and $k = 2$. The Mersenne $j = 1728$ example has $N = 2^{61}$, so its largest prime is merely 2. Conversely, the degree-83 Koblitz example has large $r$ and $k$ while its endomorphism discriminant remains $-7$.

Generic Pollard-rho work is approximately $\sqrt{r}$, or $\frac{1}{2}\log_2 r$ bits. An 83-bit field caps this scale around 41.5 bits. Extra endomorphisms can lower work further. "Large CM discriminant" in this setting is also relative: $|D_K| \le 4q$, so an 83-bit field cannot have an ordinary fundamental discriminant larger than roughly 85 bits.

05 / PRIME83

# A screened prime-order example over $2^{83} - 55$

$$p = 967140655691703339764935 3 = 2^{83} - 55, \qquad E : \; y^2 = x^3 - 3x + 154.$$

This is a useful experimental example when the desired combination is a large prime subgroup, a large embedding degree, and a large fundamental CM discriminant.

| Quantity | Exact value |
|---|---|
| Curve discriminant | $-10243584 \pmod p$ |
| $j$-invariant | $528599143799109112826988 9 \pmod p$ |
| Trace $t$ | 282648742965 |
| $N = r$ (certified prime) | 967140655691675074890 6389 |
| Cofactor | 1 |
| Pairing embedding degree | $48357032784583753744531 94 = (r-1)/2$ |
| $D_\pi = D_K = D_{\mathrm{End}}$ | $-38605735915768438953606187$ |
| Endomorphism ring | $\mathcal{O}_{\mathbb{Q}(\sqrt{-38605735915768438953606187})}$ |
| Point group | Cyclic of order $N$ |

The absolute discriminant factors as

$$13 \cdot 359 \cdot 1283 \cdot 27529 \cdot 1304411 \cdot 179548793.$$

These are distinct certified primes, and the negative discriminant is 1 modulo 4. It is therefore fundamental: $f_\pi = f_E = 1$. This determines the ordinary endomorphism ring, rather than merely guessing it from a large number.

**Quadratic twist.** The nonsquare $d = 7$ gives

$$E^{(7)} : \; y^2 = x^3 - 147x + 52822.$$

Its order is $967140655691731604639231 9 = 7r'$, where

$$r' = 13816295081310451494846 17, \qquad k' = 17270368851638064368557 7 = (r'-1)/8.$$

The twist has the same $j$, $D_\pi$, and geometric endomorphism order, but opposite trace; the two curves are *not* $\mathbb{F}_p$-isogenous. Their largest subgroup sizes are 83 and 81 bits, giving about 41.5 and 40.1 bits of generic work.

**Reproducible selection.** Test $b = 1, 2, \ldots$ in $y^2 = x^3 - 3x + b$, skipping the singular $b = 2$, and require $h \le 4$, $r \ge 2^{79}$, and a twist prime factor at least $2^{74}$. The first match is $b = 154$, after 153 nonsingular candidates. This is a transparent deterministic search, not a standardized parameter-generation process; no subgroup generator or full implementation was selected.

06 / BINARY FIELDS

# Explained clustering and a different 83-bit example

The Koblitz pair

$$K_a:\ y^2+xy=x^3+ax^2+1, \qquad a\in\{0,1\},$$

is defined over $\mathbb{F}_2$. Its degree-two Frobenius $\tau$ satisfies $\tau^2-(2a-1)\tau+2=0$, whose discriminant is already the fundamental value $-7$. Consequently

$$\mathrm{End}(K_a)=\mathcal{O}_{\mathbb{Q}(\sqrt{-7})}, \qquad D_{\mathrm{End}}=-7$$

over each extension in the report. Extension-field Frobenius is $\pi=\tau^m$ and may generate a much smaller order:

| $m$ | $f_\pi$ | $D_\pi=-7f_\pi^2$ | $h(D_\pi)$ |
|---|---|---|---|
| 53 | 68476319 | −32823043846528327 | 68476318 |
| 79 | 565842391207 | −2241243281807989416417943 | 565970327440 |
| 83 | 347450761417 | −845054221264771390935223 | 347504444820 |

The maximal order has class number one in all cases. The large class numbers above belong to the *Frobenius order*, not the actual endomorphism ring. They follow exactly from the quadratic-order class-number formula, not an uncertified numerical approximation.

For odd $m$, changing $a$ from 0 to 1 is a quadratic twist because $\mathrm{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(1)=1$. The two traces are opposite and nonzero, so the pair is geometrically isomorphic but not isogenous over its base field.

**General-coefficient controls.** Taking $b=u$ gives $j=u^{-1}$, which does not descend to $\mathbb{F}_2$. Each control pair has a large fundamental $D_\pi=D_K$ and maximal endomorphism ring, rather than $D_K=-7$. Large CM class numbers for these orders are not computed in this report.

**Screened binary83 example.** Over $\mathbb{F}_{2^{83}}$ using $f_{83}$, choose

$$y^2+xy=x^3+x^2+u^{251}.$$

Its order is $2r$, with $r=4835703278458277455699489$ (82 bits), and

$$k=805950546409712909283248, \qquad D_K=D_\pi=-384566771358516615942911871.$$

This discriminant is fundamental, so its endomorphism ring is maximal. The twist has only a 62-bit largest prime factor and cofactor 4121920. Thus its twist is materially weaker, despite the strong main subgroup.

**Another real grouping:** $b\mapsto b^2$ gives the relative Frobenius isogeny $(x,y)\mapsto(x^2,y^2)$ from $E_{a,b}$ to $E_{a,b^2}$. These curves share point counts even when their $j$ values differ. Searching $b=u^i$ therefore samples repeated Frobenius orbits; the 512 screened sides are not 512 independent random curves. Koblitz Frobenius also offers an ideal $\sqrt{m}$ rho acceleration on rational-point orbits; it is an endomorphism of the curve, not a degree-one geometric automorphism [8].

07 / EVIDENCE

# What was computed, and what remains outside scope

**Exact computation.** PARI/GP 2.17.2 counted all prime-field controls and searched examples and all binary general-coefficient examples, and computed their rational point-group structures. Every special-$j$ prime model was independently counted using its CM trace candidates and cross-checked with PARI. Binary Koblitz counts used the exact recurrence

$$T_0 = 2, \quad T_1 = 2a - 1, \quad T_m = (2a - 1)T_{m-1} - 2T_{m-2}.$$

All orders were fully factored. Lucas certificates establish primality of the reported prime factors; exact modular exponentiation establishes each embedding degree. Independent checks verified $N = q + 1 - t$, Hasse's bound, $D_\pi = f_\pi^2 D_K$, invariant factors, twist-order sums, and the certificates.

**Grouping within the sample**

| Field | Model records | Distinct traces | Supersingular records |
|---|---|---|---|
| `p61` | 10 | 9 | 2 |
| `p64` | 12 | 12 | 0 |
| `p83s` | 6 | 3 | 4 |
| `p83` | 14 | 14 | 0 |
| `b53` | 4 | 4 | 0 |
| `b79` | 4 | 4 | 0 |
| `b83` | 6 | 6 | 0 |

Distinct traces count the represented base-field isogeny classes, not every class in the field. Special-$j$ twists can include a self-isomorphic quadratic twist. Supersingular records with the same trace but different $j$ are correctly merged into one class.

**An investigation of any additional curves can follow this order:**

1. Specify the field presentation and coefficients; verify $\Delta \neq 0$ and compute $j$.
2. Count points and check Hasse; record ordinary/supersingular status and group invariant factors.
3. Factor both the curve and twist orders; identify $r, h$, and compute $k$ separately for both sides.
4. Factor $D_\pi$ into $f_\pi^2 D_K$. Prove $f_E$, or leave the actual ring unresolved when the evidence is insufficient.
5. Group by $(q, t)$ for isogeny, by $j$ for geometric isomorphism, and by $(D_K, f_E)$ for ordinary endomorphism order. Examine Frobenius orbits when searching binary coefficients.

**Limits.** The general-coefficient controls are fixed examples, not a statistically representative random sample. The searched curves satisfy stated arithmetic filters; they are not production-security recommendations. No complete class enumeration, shortest isogeny network, explicit quaternion-order basis, generic large-discriminant class-number certification, or cryptographic implementation audit was performed. NIST's B families use substantially larger binary fields; SP 800-186 also deprecates the binary families [6].

APPENDIX / EXACT DATA

# Curve-by-curve invariants

Every label below is defined on page 2. For prime rows, coefficients belong to $y^2 = x^3 + Ax + B$; for binary rows they belong to $y^2 + xy = x^3 + ax^2 + b$. All coefficients, $\Delta$, and $j$ are interpreted in the labeled field.

$D_{\mathrm{End}}$ is the ordinary endomorphism-order discriminant, or the *rational* quadratic-order discriminant on a supersingular prime-field row. Supersingular geometric rings are quaternionic. The rational point group is $\mathbb{Z}/d_1\mathbb{Z} \times \mathbb{Z}/d_2\mathbb{Z}$; a factor $d_1 = 1$ is omitted.

The companion CSV/JSON preserves these integers without PDF copy/paste ambiguities. The accompanying programs and primality certificates make the computations reproducible.

## `p61` — $q = 2305843009213693951$

**p61-E4-0** $A = -g^0,\ B = 0$ supersingular (rational ring data).
$t = 0$; $N = 2305843009213693952$; $\Delta = 64$; $j = 1728$.
$D_\pi = -9223372036854775804$; $f_\pi = 2$.
$D_K = -2305843009213693951$; $D_{\mathrm{End}} = -2305843009213693951$.
$r = 2$; $h = 1152921504606846976$; $k = 1$.
$N = 2^{61}$; group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/1152921504606846976\mathbb{Z}$.

**p61-E4-1** $A = -g^1,\ B = 0$ supersingular (rational ring data).
$t = 0$; $N = 2305843009213693952$; $\Delta = 3241792$; $j = 1728$.
$D_\pi = -9223372036854775804$; $f_\pi = 2$.
$D_K = -2305843009213693951$; $D_{\mathrm{End}} = -9223372036854775804$.
$r = 2$; $h = 1152921504606846976$; $k = 1$.
$N = 2^{61}$; group $\mathbb{Z}/2305843009213693952\mathbb{Z}$.

**p61-E3-0** $A = 0,\ B = g^0$ ordinary.
$t = -3010608196$; $N = 2305843012224302148$; $\Delta = 2305843009213693519$; $j = 0$.
$D_\pi = -159610327032401388$; $f_\pi = 230658714$.
$D_K = -3$; $D_{\mathrm{End}} = -3$.
$r = 42111239174233$; $h = 54756$; $k = 14037079724744$.
$N = 2^2 \cdot 3^4 \cdot 13^2 \cdot 42111239174233$; group $\mathbb{Z}/234\mathbb{Z} \times \mathbb{Z}/9854029966770522\mathbb{Z}$.

**p61-E3-1** $A = 0,\ B = g^1$ ordinary.
$t = -1159316027$; $N = 2305843010373009979$; $\Delta = 2305843009213102543$; $j = 0$.
$D_\pi = -7879358386395711075$; $f_\pi = 1620633455$.
$D_K = -3$; $D_{\mathrm{End}} = -3$.
$r = 2242930237$; $h = 1028049367$; $k = 2242930236$.
$N = 463 \cdot 2220409 \cdot 2242930237$; group $\mathbb{Z}/2305843010373009979\mathbb{Z}$.

**p61-E3-2** $A = 0,\ B = g^2$ ordinary.
$t = 1851292169$; $N = 2305843007362401783$; $\Delta = 2305843008404056399$; $j = 0$.
$D_\pi = -5796089341854051243$; $f_\pi = 1389974741$.
$D_K = -3$; $D_{\mathrm{End}} = -3$.
$r = 371710501$; $h = 6203330283$; $k = 123903500$.
$N = 3 \cdot 109 \cdot 163 \cdot 181 \cdot 643 \cdot 371710501$; group $\mathbb{Z}/2305843007362401783\mathbb{Z}$.

**p61-E3-3** $A = 0,\ B = g^3$ ordinary.
$t = 3010608196$; $N = 2305843006203085756$; $\Delta = 2305841900819885263$; $j = 0$.
$D_\pi = -159610327032401388$; $f_\pi = 230658714$.
$D_K = -3$; $D_{\mathrm{End}} = -3$.
$r = 157632877033$; $h = 14627932$; $k = 52544292344$.
$N = 2^2 \cdot 11^2 \cdot 30223 \cdot 157632877033$; group $\mathbb{Z}/22\mathbb{Z} \times \mathbb{Z}/104811045736503898\mathbb{Z}$.

**p61-E3-4** $A = 0,\ B = g^4$ ordinary.
$t = 1159316027$; $N = 2305843008054377925$; $\Delta = 2304325618089600079$; $j = 0$.
$D_\pi = -7879358386395711075$; $f_\pi = 1620633455$.
$D_K = -3$; $D_{\mathrm{End}} = -3$.
$r = 51131959441$; $h = 45095925$; $k = 17043986480$.
$N = 3 \cdot 5^2 \cdot 7^3 \cdot 1753 \cdot 51131959441$; group $\mathbb{Z}/35\mathbb{Z} \times \mathbb{Z}/658812288015536555\mathbb{Z}$.

**p61-E3-5** $A = 0,\ B = g^5$ ordinary.
$t = -1851292169;\ N = 2305843011064986121;\ \Delta = 228534560329183183;\ j = 0.$
$D_\pi = -5796089341854051243;\ f_\pi = 1389974741.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 53624256071278747;\ h = 43;\ k = 53624256071278746.$
$N = 43 \cdot 53624256071278747;$ group $\mathbb{Z}/230584301106498612\mathbb{1}\mathbb{Z}.$

**p61-G-0** $A = -3,\ B = 1$ ordinary.
$t = -2536646736;\ N = 2305843011750340688;\ \Delta = 1296;\ j = 2304.$
$D_\pi = -2788795373595322108;\ f_\pi = 62.$
$D_K = -725493073255807;\ D_{\mathrm{End}} = -725493073255807.$
$r = 33820067;\ h = 68179729264;\ k = 2415719.$
$N = 2^4 \cdot 16421 \cdot 259499 \cdot 33820067;$ group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/1152921505875170344\mathbb{Z}.$

**p61-G-1** $A = -4107,\ B = 50653$ ordinary.
$t = 2536646736;\ N = 2305843006677047216;\ \Delta = 3325181426064;\ j = 2304.$
$D_\pi = -2788795373595322108;\ f_\pi = 62.$
$D_K = -725493073255807;\ D_{\mathrm{End}} = -725493073255807.$
$r = 3417821983;\ h = 674652752;\ k = 379757998.$
$N = 2^4 \cdot 17 \cdot 29 \cdot 31^2 \cdot 89 \cdot 3417821983;$ group $\mathbb{Z}/62\mathbb{Z} \times \mathbb{Z}/37191016236726568\mathbb{Z}.$

## p64 — $q = 18446744069414584321$

**p64-E4-0** $A = -g^0,\ B = 0$ ordinary.
$t = -8589934590;\ N = 18446744078004518912;\ \Delta = 64;\ j = 1728.$
$D_\pi = -17179869184;\ f_\pi = 65536.$
$D_K = -4;\ D_{\mathrm{End}} = -4.$
$r = 6700417;\ h = 2753074036736;\ k = 558368.$
$N = 2^{32} \cdot 641 \cdot 6700417;$ group $\mathbb{Z}/65536\mathbb{Z} \times \mathbb{Z}/281474976776192\mathbb{Z}.$

**p64-E4-1** $A = -g^1,\ B = 0$ ordinary.
$t = 131072;\ N = 18446744069414453250;\ \Delta = 21952;\ j = 1728.$
$D_\pi = -73786976260478468100;\ f_\pi = 4294967295.$
$D_K = -4;\ D_{\mathrm{End}} = -4.$
$r = 49477;\ h = 372834732692250;\ k = 16492.$
$N = 2 \cdot 3^2 \cdot 5^3 \cdot 17^2 \cdot 257^2 \cdot 49477 \cdot 8681;$ group $\mathbb{Z}/65535\mathbb{Z} \times \mathbb{Z}/281479271677950\mathbb{Z}.$

**p64-E4-2** $A = -g^2,\ B = 0$ ordinary.
$t = 8589934590;\ N = 18446744060824649732;\ \Delta = 7529536;\ j = 1728.$
$D_\pi = -17179869184;\ f_\pi = 65536.$
$D_K = -4;\ D_{\mathrm{End}} = -4.$
$r = 2393421905249;\ h = 7707268;\ k = 598355476312.$
$N = 2^2 \cdot 281 \cdot 6857 \cdot 2393421905249;$ group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/9223372030412324866\mathbb{Z}.$

**p64-E4-3** $A = -g^3,\ B = 0$ ordinary.
$t = -131072;\ N = 18446744069414715394;\ \Delta = 2582630848;\ j = 1728.$
$D_\pi = -73786976260478468100;\ f_\pi = 4294967295.$
$D_K = -4;\ D_{\mathrm{End}} = -4.$
$r = 384773;\ h = 47941887994778;\ k = 384772.$
$N = 2 \cdot 5581 \cdot 65537^2 \cdot 384773;$ group $\mathbb{Z}/65537\mathbb{Z} \times \mathbb{Z}/281470681743362\mathbb{Z}.$

**p64-E3-0** $A = 0,\ B = g^0$ ordinary.
$t = 4294967294;\ N = 18446744065119617028;\ \Delta = 18446744069414583889;\ j = 0.$
$D_\pi = -55340232221128654848;\ f_\pi = 4294967296.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 2903110321;\ h = 6354131268;\ k = 50931760.$
$N = 2^2 \cdot 3 \cdot 529510939 \cdot 2903110321;$ group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/9223372032559808514\mathbb{Z}.$

**p64-E3-1** $A = 0,\ B = g^1$ ordinary.
$t = 8589934591;\ N = 18446744060824649731;\ \Delta = 18446744069414563153;\ j = 0.$
$D_\pi = -3;\ f_\pi = 1.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 318032437;\ h = 58002712663;\ k = 106010812.$
$N = 7 \cdot 7417 \cdot 1117177 \cdot 318032437;$ group $\mathbb{Z}/184467440608246497\mathbb{3}\mathbb{1}\mathbb{Z}.$

**p64-E3-2** $A = 0,\ B = g^2$ ordinary.
$t = 4294967297;\ N = 18446744065119617025;\ \Delta = 18446744069413547089;\ j = 0.$
$D_\pi = -55340232195358851075;\ f_\pi = 4294967295.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 65537;\ h = 281470681677825;\ k = 1.$
$N = 3^2 \cdot 5^2 \cdot 17^2 \cdot 257^2 \cdot 65537^2;$ group $\mathbb{Z}/4294967295\mathbb{Z} \times \mathbb{Z}/4294967295\mathbb{Z}.$

**p64-E3-3** $A = 0,\ B = g^3$ ordinary.
$t = -4294967294;\ N = 18446744073709551616;\ \Delta = 18446744069363759953;\ j = 0.$
$D_\pi = -55340232221128654848;\ f_\pi = 4294967296.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 2;\ h = 9223372036854775808;\ k = 1.$
$N = 2^{64};$ group $\mathbb{Z}/4294967296\mathbb{Z} \times \mathbb{Z}/4294967296\mathbb{Z}.$

**p64-E3-4** $A = 0,\ B = g^4$ ordinary.
$t = -8589934591;\ N = 18446744078004518913;\ \Delta = 18446744066924190289;\ j = 0.$
$D_\pi = -3;\ f_\pi = 1.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 22253377;\ h = 828941336769;\ k = 32.$
$N = 3 \cdot 7 \cdot 13 \cdot 97 \cdot 193 \cdot 241 \cdot 673 \cdot 22253377;$ group $\mathbb{Z}/18446744078004518913\mathbb{Z}.$

**p64-E3-5** $A = 0,\ B = g^5$ ordinary.
$t = -4294967297;\ N = 18446744073709551619;\ \Delta = 18446743947385276753;\ j = 0.$
$D_\pi = -55340232195358851075;\ f_\pi = 4294967295.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 39463029637;\ h = 467443687;\ k = 636500478.$
$N = 467443687 \cdot 39463029637;$ group $\mathbb{Z}/18446744073709551619\mathbb{Z}.$

**p64-G-0** $A = -3,\ B = 1$ ordinary.
$t = -7816789861;\ N = 18446744077231374183;\ \Delta = 1296;\ j = 2304.$
$D_\pi = -12684772546625937963;\ f_\pi = 1.$
$D_K = -12684772546625937963;\ D_{\mathrm{End}} = -12684772546625937963.$
$r = 311886008947;\ h = 59145789;\ k = 311886008946.$
$N = 3 \cdot 59 \cdot 334157 \cdot 311886008947;$ group $\mathbb{Z}/18446744077231374183\mathbb{Z}.$

**p64-G-1** $A = -147,\ B = 343$ ordinary.
$t = 7816789861;\ N = 18446744061597794461;\ \Delta = 152473104;\ j = 2304.$
$D_\pi = -12684772546625937963;\ f_\pi = 1.$
$D_K = -12684772546625937963;\ D_{\mathrm{End}} = -12684772546625937963.$
$r = 214844272272601;\ h = 85861;\ k = 26855534034075.$
$N = 19 \cdot 4519 \cdot 214844272272601;$ group $\mathbb{Z}/18446744061597794461\mathbb{Z}.$

## p83s — $q = 967140598045628109422591 9$

**p83s-E4-0** $A = -g^0,\ B = 0$ supersingular (rational ring data).
$t = 0;\ N = 96714059804562810942259 20;\ \Delta = 64;\ j = 1728.$
$D_\pi = -38685623921825124376903676;\ f_\pi = 2.$
$D_K = -96714059804562810942259 19;\ D_{\mathrm{End}} = -96714059804562810942259 19.$
$r = 241;\ h = 40130315271602826117120;\ k = 2.$
$N = 2^{59} \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241;$ group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/48357029902281405471129 60\mathbb{Z}.$

**p83s-E4-1** $A = -g^1,\ B = 0$ supersingular (rational ring data).
$t = 0;\ N = 96714059804562810942259 20;\ \Delta = 85184;\ j = 1728.$
$D_\pi = -38685623921825124376903676;\ f_\pi = 2.$
$D_K = -96714059804562810942259 19;\ D_{\mathrm{End}} = -38685623921825124376903676.$
$r = 241;\ h = 40130315271602826117120;\ k = 2.$
$N = 2^{59} \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241;$ group $\mathbb{Z}/9671405980456281094225920\mathbb{Z}.$

**p83s-E3-0** $A = 0,\ B = g^0$ supersingular (rational ring data).
$t = 0;\ N = 96714059804562810942259 20;\ \Delta = 96714059804562810942254 87;\ j = 0.$
$D_\pi = -38685623921825124376903676;\ f_\pi = 2.$
$D_K = -96714059804562810942259 19;\ D_{\mathrm{End}} = -38685623921825124376903676.$
$r = 241;\ h = 40130315271602826117120;\ k = 2.$
$N = 2^{59} \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241;$ group $\mathbb{Z}/9671405980456281094225920\mathbb{Z}.$

**p83s-E3-1** $A = 0,\ B = g^{1}$ supersingular (rational ring data).
$t = 0$; $N = 96714059804562810942259 20$; $\Delta = 96714059804562810941736 47$; $j = 0$.
$D_\pi = -38685623921825124376903676$; $f_\pi = 2$.
$D_K = -96714059804562810942259 19$; $D_{\mathrm{End}} = -38685623921825124376903676$.
$r = 241$; $h = 4013031527160282611720$; $k = 2$.
$N = 2^{59} \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241$; group $\mathbb{Z}/967140598045628109422592 0\mathbb{Z}$.

**p83s-G-0** $A = -3,\ B = 1$ ordinary.
$t = 423089571240$; $N = 96714059804558580046546 80$; $\Delta = 1296$; $j = 2304$.
$D_\pi = -38506619136533077341766076$; $f_\pi = 2$.
$D_K = -96266547841332693353441519$; $D_{\mathrm{End}} = -96266547841332693353441519$.
$r = 10609823198306767$; $h = 911552040$; $k = 1768303866384461$.
$N = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 361727 \cdot 10609823198306767$; group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/48357029902279290023273 40\mathbb{Z}$.

**p83s-G-1** $A = -363,\ B = 1331$ ordinary.
$t = -423089571240$; $N = 96714059804567041837971 60$; $\Delta = 2295943056$; $j = 2304$.
$D_\pi = -38506619136533077341766076$; $f_\pi = 2$.
$D_K = -96266547841332693353441519$; $D_{\mathrm{End}} = -96266547841332693353441519$.
$r = 710774671$; $h = 13606852319103960$; $k = 355387335$.
$N = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 9549499 \cdot 565427 \cdot 710774671$; group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/48357029902283520918985 80\mathbb{Z}$.

## p83 — $q = 96714065569170333976493 53$

**p83-E4-0** $A = -g^{0},\ B = 0$ ordinary.
$t = -4516048014294$; $N = 96714065569215494456636 48$; $\Delta = 64$; $j = 1728$.
$D_\pi = -18290936560259353162278976$; $f_\pi = 2138395225412$.
$D_K = -4$; $D_{\mathrm{End}} = -4$.
$r = 2523639968257$; $h = 3832324213664$; $k = 210303330688$.
$N = 2^5 \cdot 29 \cdot 73 \cdot 149 \cdot 313 \cdot 1213 \cdot 2523639968257$; group $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/24178516392303873614159 12\mathbb{Z}$.

**p83-E4-1** $A = -g^{1},\ B = 0$ ordinary.
$t = 4276790450824$; $N = 96714065569127566071985 30$; $\Delta = 21952$; $j = 1728$.
$D_\pi = -20394689667408780428318436$; $f_\pi = 2258024007147$.
$D_K = -4$; $D_{\mathrm{End}} = -4$.
$r = 3298407032069$; $h = 2932144657370$; $k = 3298407032068$.
$N = 2 \cdot 5 \cdot 3298407032069 \cdot 293214465737$; group $\mathbb{Z}/96714065569127566071985 30\mathbb{Z}$.

**p83-E4-2** $A = -g^{2},\ B = 0$ ordinary.
$t = 4516048014294$; $N = 96714065569125173496350 60$; $\Delta = 7529536$; $j = 1728$.
$D_\pi = -18290936560259353162278976$; $f_\pi = 2138395225412$.
$D_K = -4$; $D_{\mathrm{End}} = -4$.
$r = 48357032784562586748175 30$; $h = 20$; $k = 80595054640937644580292$.
$N = 2^2 \cdot 5 \cdot 48357032784562586748175 30$; group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/48357032784562586748175 30\mathbb{Z}$.

**p83-E4-3** $A = -g^{3},\ B = 0$ ordinary.
$t = -4276790450824$; $N = 96714065569213101881001 78$; $\Delta = 2582630848$; $j = 1728$.
$D_\pi = -20394689667408780428318436$; $f_\pi = 2258024007147$.
$D_K = -4$; $D_{\mathrm{End}} = -4$.
$r = 238906342495956479129$; $h = 40482$; $k = 119453171247978239564$.
$N = 2 \cdot 3^2 \cdot 13 \cdot 173 \cdot 238906342495956479129$; group $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/32238021856404367293667 26\mathbb{Z}$.

**p83-E3-0** $A = 0,\ B = g^{0}$ ordinary.
$t = -785790761722$; $N = 96714065569178191884110 76$; $\Delta = 96714065569170333976489 21$; $j = 0$.
$D_\pi = -38068159106460492610192128$; $f_\pi = 3562216496624$.
$D_K = -3$; $D_{\mathrm{End}} = -3$.
$r = 648204994840489$; $h = 14920290084$; $k = 324102497420244$.
$N = 2^2 \cdot 3 \cdot 7^3 \cdot 3624949 \cdot 648204994840489$; group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/48357032784589095942055 38\mathbb{Z}$.

**p83-E3-1** $A = 0,\ B = g^{1}$ ordinary.
$t = -5736220125797$; $N = 96714065569227696177751 51$; $\Delta = 96714065569170333976281 85$; $j = 0$.
$D_\pi = -57814048960695830857122 03$; $f_\pi = 1388212867451$.
$D_K = -3$; $D_{\mathrm{End}} = -3$.
$r = 271647827404363$; $h = 35602738477$; $k = 90549275801454$.
$N = 97 \cdot 367038541 \cdot 271647827404363$; group $\mathbb{Z}/96714065569227696177751 51\mathbb{Z}$.

**p83-E3-2** $A = 0,\ B = g^2$ ordinary.
$t = -4950429364075;\ N = 967140655692198382701342 9;\ \Delta = 96714065569170333966121;\ j = 0.$
$D_\pi = -14178875338972124689991787;\ f_\pi = 2174003629173.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 5957986933;\ h = 1623267500530113;\ k = 1489496733.$
$N = 3^2 \cdot 137623 \cdot 1310558959 \cdot 5957986933;$ group $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/322380218564066127567114 3\mathbb{Z}.$

**p83-E3-3** $A = 0,\ B = g^3$ ordinary.
$t = 785790761722;\ N = 967140655691624760688763 2;\ \Delta = 96714065569170333468249 85;\ j = 0.$
$D_\pi = -38068159106460492610192128;\ f_\pi = 3562216496624.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 148553185010387189833;\ h = 65104;\ k = 14855318501038718983 2.$
$N = 2^4 \cdot 13 \cdot 313 \cdot 148553185010387189833;$ group $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/241785163922906190172190 8\mathbb{Z}.$

**p83-E3-4** $A = 0,\ B = g^4$ ordinary.
$t = 5736220125797;\ N = 967140655691129717752355 7;\ \Delta = 96714065569170309072553 21;\ j = 0.$
$D_\pi = -57814048960695830857122 03;\ f_\pi = 1388212867451.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 24290441341356013;\ h = 398156889;\ k = 6072610335339003.$
$N = 3 \cdot 13 \cdot 37 \cdot 275923 \cdot 24290441341356013;$ group $\mathbb{Z}/9671406556911297177523557\mathbb{Z}.$

**p83-E3-5** $A = 0,\ B = g^5$ ordinary.
$t = 4950429364075;\ N = 967140655691208296828527 9;\ \Delta = 96714065569169113683417 85;\ j = 0.$
$D_\pi = -14178875338972124689991787;\ f_\pi = 2174003629173.$
$D_K = -3;\ D_{\mathrm{End}} = -3.$
$r = 34617830743;\ h = 279376447031353;\ k = 1049025174.$
$N = 7^2 \cdot 151 \cdot 10099 \cdot 3738853 \cdot 34617830743;$ group $\mathbb{Z}/9671406556912082968285279\mathbb{Z}.$

**p83-G-0** $A = -3,\ B = 1$ ordinary.
$t = 1299209545187;\ N = 967140655691573418810416 7;\ \Delta = 1296;\ j = 2304.$
$D_\pi = -3699768078536312219573244 3;\ f_\pi = 1.$
$D_K = -3699768078536312219573244 3;\ D_{\mathrm{End}} = -3699768078536312219573244 3.$
$r = 3223802185638578062701389;\ h = 3;\ k = 805950546409644515675347.$
$N = 3 \cdot 3223802185638578062701389;$ group $\mathbb{Z}/9671406556915734188104167\mathbb{Z}.$

**p83-G-1** $A = -147,\ B = 343$ ordinary.
$t = -1299209545187;\ N = 967140655691833260719454 1;\ \Delta = 152473104;\ j = 2304.$
$D_\pi = -3699768078536312219573244 3;\ f_\pi = 1.$
$D_K = -3699768078536312219573244 3;\ D_{\mathrm{End}} = -3699768078536312219573244 3.$
$r = 48878803567519;\ h = 197865042739;\ k = 24439401783759.$
$N = 7 \cdot 13 \cdot 271 \cdot 797 \cdot 10067 \cdot 48878803567519;$ group $\mathbb{Z}/9671406556918332607194541\mathbb{Z}.$

**p83-S-0** $A = -3,\ B = 154$ ordinary.
$t = 282648742965;\ N = 967140655691675074890638 9;\ \Delta = 96714065569170333874057 69;$
$j = 5285991437991091128269889.$
$D_\pi = -3860573591576843895360618 7;\ f_\pi = 1.$
$D_K = -3860573591576843895360618 7;\ D_{\mathrm{End}} = -3860573591576843895360618 7.$
$r = 96714065569167507489063 89;\ h = 1;\ k = 4835703278458375374453194.$
$N = 96714065569167507489063 89;$ group $\mathbb{Z}/9671406556916750748906389\mathbb{Z}.$

**p83-S-1** $A = -147,\ B = 52822$ ordinary.
$t = -282648742965;\ N = 967140655691731604639231 9;\ \Delta = 96714065569158282502353 37;$
$j = 5285991437991091128269889.$
$D_\pi = -3860573591576843895360618 7;\ f_\pi = 1.$
$D_K = -3860573591576843895360618 7;\ D_{\mathrm{End}} = -3860573591576843895360618 7.$
$r = 1381629508131045149484617;\ h = 7;\ k = 172703688516380643685577.$
$N = 7 \cdot 1381629508131045149484617;$ group $\mathbb{Z}/9671406556917316046392319\mathbb{Z}.$

## b53 — $q = 9007199254740992$

**K53-a0** $a = 0,\ b = 1$ ordinary.
$t = -56619371;\ N = 9007199311360364;\ \Delta = 1;\ j = 1.$
$D_\pi = -32823043846528327;\ f_\pi = 68476319.$
$D_K = -7;\ D_{\mathrm{End}} = -7.$
$r = 21044858204113;\ h = 428;\ k = 66178799384.$
$N = 2^2 \cdot 107 \cdot 21044858204113;$ group $\mathbb{Z}/9007199311360364\mathbb{Z}.$

**K53-a1** $a = 1,\ b = 1$ ordinary.
$t = 56619371;\ N = 9007199198121622;\ \Delta = 1;\ j = 1.$
$D_\pi = -32823043846528327;\ f_\pi = 68476319.$
$D_K = -7;\ D_{\mathrm{End}} = -7.$
$r = 12066511;\ h = 746462602;\ k = 113835.$
$N = 2 \cdot 107 \cdot 3488143 \cdot 12066511;$ group $\mathbb{Z}/9007199198121622\mathbb{Z}.$

**B53-a0** $a = 0,\ b = u$ ordinary.
$t = 120448001;\ N = 9007199134292992;\ \Delta = u;\ j = u^{-1}.$
$D_\pi = -21521076074067967;\ f_\pi = 1.$
$D_K = -21521076074067967;\ D_{\mathrm{End}} = -21521076074067967.$
$r = 20036658097;\ h = 449536;\ k = 1252291131.$
$N = 2^{10} \cdot 439 \cdot 20036658097;$ group $\mathbb{Z}/9007199134292992\mathbb{Z}.$

**B53-a1** $a = 1,\ b = u$ ordinary.
$t = -120448001;\ N = 9007199375188994;\ \Delta = u;\ j = u^{-1}.$
$D_\pi = -21521076074067967;\ f_\pi = 1.$
$D_K = -21521076074067967;\ D_{\mathrm{End}} = -21521076074067967.$
$r = 112052167;\ h = 80383982;\ k = 56026083.$
$N = 2 \cdot 7 \cdot 5741713 \cdot 112052167;$ group $\mathbb{Z}/9007199375188994\mathbb{Z}.$

## b79 — $q = 6044629098073145873530 88$

**K79-a0** $a = 0,\ b = 1$ ordinary.
$t = -420247971347;\ N = 6044629098077348353244 36;\ \Delta = 1;\ j = 1.$
$D_\pi = -2241243281807989416417943;\ f_\pi = 565842391207.$
$D_K = -7;\ D_{\mathrm{End}} = -7.$
$r = 14377452373;\ h = 42042421294532;\ k = 181993068.$
$N = 2^2 \cdot 149627 \cdot 14377452373 \cdot 70245379;$ group $\mathbb{Z}/6044629098077348353244 36\mathbb{Z}.$

**K79-a1** $a = 1,\ b = 1$ ordinary.
$t = 420247971347;\ N = 6044629098068943393817 42;\ \Delta = 1;\ j = 1.$
$D_\pi = -2241243281807989416417943;\ f_\pi = 565842391207.$
$D_K = -7;\ D_{\mathrm{End}} = -7.$
$r = 953411529663871197763;\ h = 634;\ k = 12068500375492040478.$
$N = 2 \cdot 317 \cdot 953411529663871197763;$ group $\mathbb{Z}/6044629098068943393817 42\mathbb{Z}.$

**B79-a0** $a = 0,\ b = u$ ordinary.
$t = 117766930433;\ N = 6044629098071968204226 56;\ \Delta = u;\ j = u^{-1}.$
$D_\pi = -2403982589325647287844863;\ f_\pi = 1.$
$D_K = -2403982589325647287844863;\ D_{\mathrm{End}} = -2403982589325647287844863.$
$r = 7159613457677453;\ h = 84426752;\ k = 7159613457677452.$
$N = 2^{14} \cdot 5153 \cdot 7159613457677453;$ group $\mathbb{Z}/6044629098071968204226 56\mathbb{Z}.$

**B79-a1** $a = 1,\ b = u$ ordinary.
$t = -117766930433;\ N = 6044629098074323542835 22;\ \Delta = u;\ j = u^{-1}.$
$D_\pi = -2403982589325647287844863;\ f_\pi = 1.$
$D_K = -2403982589325647287844863;\ D_{\mathrm{End}} = -2403982589325647287844863.$
$r = 1218884333772037;\ h = 495914906;\ k = 1218884333772036.$
$N = 2 \cdot 29 \cdot 307 \cdot 27851 \cdot 1218884333772037;$ group $\mathbb{Z}/6044629098074323542835 22\mathbb{Z}.$

## b83 — $q = 9671406556917033397649408$

**K83-a0** $a = 0,\ b = 1$ ordinary.
$t = -6151469093347;\ N = 9671406556923184866742756;\ \Delta = 1;\ j = 1.$
$D_\pi = -845054221264771390935223;\ f_\pi = 347450761417.$
$D_K = -7;\ D_{\mathrm{End}} = -7.$
$r = 2417851639230796216685689;\ h = 4;\ k = 14565371320667447088468.$
$N = 2^2 \cdot 2417851639230796216685689;$ group $\mathbb{Z}/9671406556923184866742756\mathbb{Z}.$

**K83-a1** $a = 1,\ b = 1$ ordinary.
$t = 6151469093347;\ N = 96714065569108819285560 62;\ \Delta = 1;\ j = 1.$
$D_\pi = -845054221264771390935223;\ f_\pi = 347450761417.$
$D_K = -7;\ D_{\mathrm{End}} = -7.$
$r = 8569786107849059;\ h = 1128547018;\ k = 103250435034326.$
$N = 2 \cdot 10459 \cdot 53951 \cdot 8569786107849059;$ group $\mathbb{Z}/967140655691088192855606 2\mathbb{Z}.$

**B83-a0** $a = 0,\ b = u$ ordinary.
$t = -5450226835455;\ N = 96714065569224836244848 64;\ \Delta = u;\ j = u^{-1}.$
$D_\pi = -898065366975430994554060 7;\ f_\pi = 1.$
$D_K = -898065366975430994554060 7;\ D_{\mathrm{End}} = -898065366975430994554060 7.$
$r = 34495408063;\ h = 280367941705728;\ k = 17247704031.$
$N = 2^{13} \cdot 3 \cdot 11408200753 \cdot 34495408063;$ group $\mathbb{Z}/967140655692248362448486 4\mathbb{Z}.$

**B83-a1** $a = 1,\ b = u$ ordinary.
$t = 5450226835455;\ N = 96714065569115831708139 54;\ \Delta = u;\ j = u^{-1}.$
$D_\pi = -898065366975430994554060 7;\ f_\pi = 1.$
$D_K = -898065366975430994554060 7;\ D_{\mathrm{End}} = -898065366975430994554060 7.$
$r = 19679320883;\ h = 491450218958838;\ k = 1513793914.$
$N = 2 \cdot 3 \cdot 293 \cdot 236699 \cdot 1181039 \cdot 19679320883;$ group $\mathbb{Z}/967140655691158317081395 4\mathbb{Z}.$

**S83-a0** $a = 0,\ b = u^{251}$ ordinary.
$t = -478486250431;\ N = 96714065569175118838998 40;\ \Delta = u^{251};\ j = u^{-251}.$
$D_\pi = -384566771358166159429118 71;\ f_\pi = 1.$
$D_K = -384566771358166159429118 71;\ D_{\mathrm{End}} = -384566771358166159429118 71.$
$r = 2346335338123377427;\ h = 4121920;\ k = 782111779374459142.$
$N = 2^6 \cdot 5 \cdot 11 \cdot 1171 \cdot 2346335338123377427;$ group $\mathbb{Z}/967140655691751188389984 0\mathbb{Z}.$

**S83-a1** $a = 1,\ b = u^{251}$ ordinary.
$t = 478486250431;\ N = 96714065569165549113989 78;\ \Delta = u^{251};\ j = u^{-251}.$
$D_\pi = -384566771358166159429118 71;\ f_\pi = 1.$
$D_K = -384566771358166159429118 71;\ D_{\mathrm{End}} = -384566771358166159429118 71.$
$r = 4835703278458277455699489;\ h = 2;\ k = 805950546409712909283248.$
$N = 2 \cdot 4835703278458277455699489;$ group $\mathbb{Z}/967140655691655491139897 8\mathbb{Z}.$

REFERENCES / REPRODUCIBILITY

# Sources and the calculation bundle

These references support the classification framework; the numerical tables are original computations from the specified equations. References are hyperlinked.

[1] Sam Lichtenstein, *Tate's isogeny theorem for abelian varieties over finite fields*. Stanford seminar notes, hosted by Brian Conrad. https://virtualmath1.stanford.edu/~conrad/mordellsem/Notes/L03.pdf

[2] G. Bisson and A. V. Sutherland, *Computing the endomorphism ring of an ordinary elliptic curve over a finite field*, Journal of Number Theory (2011), 815–831. https://arxiv.org/abs/0902.4670

[3] A. V. Sutherland, *Isogeny volcanoes*. https://arxiv.org/abs/1208.5370

[4] Standards for Efficient Cryptography Group, *SEC 1: Elliptic Curve Cryptography*, version 2.0 (2009). https://www.secg.org/sec1-v2.pdf

[5] PARI/GP, *Catalogue of functions: elliptic curves*, especially `ellcard`, `ellgroup`, and `ellissupersingular`. https://pari.math.u-bordeaux.fr/dochtml/html/Elliptic_curves.html

[6] NIST, *SP 800-186: Recommendations for Discrete Logarithm-Based Cryptography: Elliptic Curve Domain Parameters* (2023). https://csrc.nist.gov/pubs/sp/800/186/final

[7] IRTF, *RFC 7748: Elliptic Curves for Security* (2016), Appendix A. Example generation criteria treating MOV degree and CM discriminant separately. https://www.rfc-editor.org/rfc/rfc7748.html

[8] T. Oliveira, J. López, D. Cervantes-Vázquez, and F. Rodríguez-Henríquez, *Koblitz curves over quadratic fields*, IACR ePrint 2016/603. Reference for Frobenius-orbit acceleration; no measured attack timings are claimed here. https://eprint.iacr.org/2016/603

### Files included in the calculation bundle

- `prime_compute.py`, `binary_compute.py`: point counts, factorization, embedding degrees, ring arguments, and deterministic searches.
- `prime_data.json`, `binary_data.json`: original detailed results, including proof details and search metadata.
- `prime_certificates.json`, `binary_certificates.json`: Lucas primality certificates.
- `normalized_data.json`, `curve_invariants.csv`: a unified 56-row comparison.
- `verify_report.py`: independent arithmetic/certificate verification.
- `build_report.py`, the generated `.tex`, and the standalone comparison figure: report-generation sources.

**Requirements.** Python 3 with SymPy; PARI/GP for point counts and group structure. Report regeneration also uses Matplotlib and XeLaTeX. Set `CURVE_REPORT_GP` to a GP executable when it is not on the executable search path. Run the two calculation programs, then `normalize.py`, `verify_report.py`, and `build_report.py`; compile the generated `elliptic_curve_fields_report.tex` with XeLaTeX.

All prime claims in the reported factorizations have certificates. The code separately checks the embedding-order minimality, rather than only verifying that an exponent happens to work. No certified large generic class numbers are reported.

# A golden EC benchmark battery

Updated experimental scope • release 1.0.0 • 12 October 2026

The original field-and-curve report is extended here with executable polynomial relation-search fixtures, measured solver results, operational isogenies, two-term endomorphism decompositions, and a separate genuine GLV/GLS four-term example. This addendum gives the current artifact counts and measurement contract; these counts supersede any earlier vector inventory.

| Layer | Current content | Purpose |
|---|---|---|
| Arithmetic corpus | 7 fields; 56 curves; 6,931 known answers | Arithmetic, encoding, rejection, subgroup membership, maps |
| Polynomial battery | 69 fixtures; main battery 55 (49 binary, 6 prime) | Two-factor Semaev relation solving with exact finite domains |
| GLV2 / Frobenius | 30 supported of 56; 480 scalar vectors | Map stability, eigenvalues, lattices, signed reconstruction |
| GLV + GLS4 supplement | 82-bit quadratic field; 80-bit subgroup | Verified four-term lattice with 20-bit component bound |

## Which curves can be grouped?

Over each fixed field, equal Frobenius trace characterizes an isogeny class; the 56 represented models occupy 52 such classes. Equal j-invariant groups geometric isomorphism classes and twists, and is a different partition. An ordinary endomorphism-order discriminant records further structure inside an isogeny class. None of these labels determines the performance of a particular polynomial presentation.

## Fields retained from the original report

Prime fields: $2^{61}-1$, $2^{64}-2^{32}+1$, $2^{83}-2^{59}-1$, and $2^{83}-55$. Binary fields: degrees 53, 79, and 83 with the explicit irreducible polynomial bases in the dataset. The binary examples are custom small-field models analogous in shape to NIST binary curves, rather than NIST standardized parameters.

A separate 14-fixture coverage extension adds degree-53 and degree-79 binary cases plus the other three prime fields. The combined 69 polynomial fixtures cover all seven core fields; the main complexity battery retains its original 55 cases and measurements.

These small fields are experimental benchmarks. An approximately 80-bit prime subgroup admits generic attacks of roughly $2^{40}$ group operations; this corpus makes no production-security recommendation.

# The polynomial experiments

Index-calculus relation finding, with fixed two-summand targets

Each case seeks actual selected-subgroup points P and Q satisfying P+Q=T. The target is fixed before solving. Its x-coordinate is substituted into $f_3(x_1,x_2,x_3)$, and all coefficients, variables, monomial order, finite domains, and preprocessing choices are recorded as canonical sparse input.

## Semaev equations used

For $y^2=x^3+Ax+B$ in odd characteristic:

$$f_3 = (x_1-x_2)^2\, x_3^2 - 2[(x_1+x_2)(x_1x_2+A)+2B]\, x_3 + (x_1x_2-A)^2 - 4B(x_1+x_2)$$

For $y^2+xy=x^3+ax^2+b$ in characteristic two:

$$f_3 = (x_1x_2+x_1x_3+x_2x_3)^2 + x_1x_2x_3 + b$$

The binary system expands every extension-field coordinate over $F_2$. A factor-base subspace of dimension d produces 2d Boolean variables. Semaev-derived equations are Boolean-normalized, while $v^2-v=0$ field equations remain explicit. With a fixed target the normalized $f_3$ coordinates have degree at most two; subgroup-membership indicators can have higher degree and must be retained in the input-degree statistics.

## Main 55-fixture complexity grid

| Cases | Factor bases and target cohorts |
|---|---|
| 49 binary | K83-a0, S83-a1, B83-a0; low monomials d=2–6 and recorded subgroup-seeded linear bases d=2–5 |
| 6 prime | p83-S-0; 4, 8, or 16 factor-base x values from small generator multiples; exact split membership polynomials |
| 25 planted | Constructed from eligible factor-base points; exercise recovery, not random-target success probability |
| 30 seeded subgroup | Independently sampled reproducible nonzero subgroup targets; includes 5 empty-factor-base diagnostics |

Low-monomial B83-a0 bases are empty after the selected-subgroup filter at every tested d=2–6. They remain useful unit-ideal diagnostics. Subgroup-seeded bases explicitly include eligible generator x values; their construction is biased and has its own result stratum.

A polynomial root is only a candidate. The runner lifts it to the recorded rational points, verifies membership in the chosen cyclic subgroup, and checks the exact signed sum T. It retains ordered signed decompositions and their swap orbits, allowing repeated factors. Characteristic-two polynomial coefficients do not turn −P into P in the odd-order subgroup.

The 14 added coverage fixtures use K53-a1 and K79-a1 with seeded dimension-2/3 bases, and p61-G-0, p64-G-0, and p83s-G-0 with four-x-value bases. Each configuration has a planted and a sampled target. They extend field coverage rather than the complexity grid.

# What is measured

Separate yield, presentation degree, solver work, and elapsed time

| Quantity | Definition in this battery |
|---|---|
| Target success | Targets with at least one verified relation / all attempted targets; a lower bound when solves are censored |
| Relations per attempt | Verified distinct unordered signed decompositions / all attempted targets; may exceed one |
| Final basis degree | Maximum total degree among polynomials in the computed final Gröbner basis |
| Degree of regularity | First d where the degree-d component of S/J is zero; J is generated by the highest-degree homogeneous parts of the exact recorded affine inputs |
| Solving degree | Maximum degree reached during the solver computation; unmeasured (null), because SymPy does not expose it |
| Solve time | Prebuilt-equation parsing/setup + Gröbner basis + finite-domain root extraction + independent subgroup/point verification |

## Timing boundary

The runs solve prebuilt fixtures. Factor-base and target construction, Semaev construction, Weil restriction, and golden-truth enumeration occur before the benchmark. Their generation time was not recorded. Both fixture_generation_seconds and generation_inclusive_total_pipeline_seconds are null; the reported solve time is not a generation-inclusive attack time.

A separate observed subprocess wall interval includes Python/SymPy startup and the optional homogeneous-ideal regularity diagnostic. Each case has a 2.5-second subprocess budget. A completed solve remains completed if only its later diagnostic times out. An unfinished solve is censored, with unknown relation yield, rather than an empty mathematical relation set.

## Why the degree labels matter

Regularity here is an invariant of the recorded generator presentation. Membership filters, field equations, and preprocessing affect J. An affine unit ideal can therefore have final basis degree zero and a positive homogeneous-input regularity. Neither number measures the maximum intermediate degree reached by the solver.

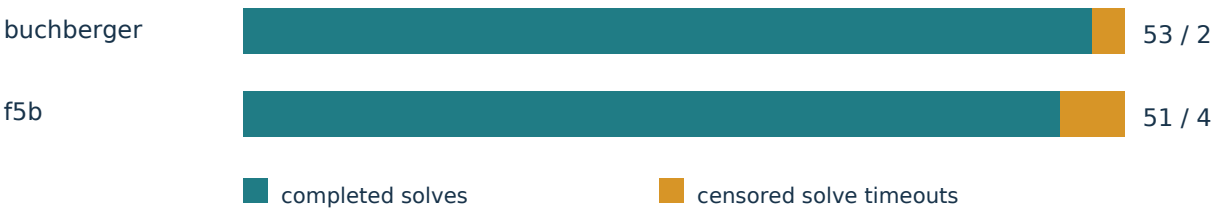
| Buchberger example | Input degree | Final GB degree | Regularity |
|---|---|---|---|
| K83-a0, low d=2, planted | 2 | 1 | 2 |
| S83-a1, low d=5, planted | 4 | 2 | 6 |
| p83-S-0, base 4, planted | 4 | 3 | 5 |
| p83-S-0, base 4, sampled | 4 | 0 (unit ideal) | 5 |

Roots are extracted by evaluating the computed basis over the declared finite grid. The baseline uses actual Gröbner computation, followed by an explicit enumeration stage; it makes no fast triangular/FGLM-root-solver claim. Golden answers are compared only after solver roots have been obtained.

# Measured solver baseline

SymPy 1.14.0 • grevlex • 2.5 seconds per case • Linux x86_64, Python 3.12.14

| Method | Complete / 55 | Solve timeouts | Diagnostic-only timeouts | Suite wall (s) | Σ solved (s) |
|---|---|---|---|---|---|
| buchberger | 53 | 2 | 1 | 45.84 | 19.71 |
| f5b | 51 | 4 | 10 | 60.69 | 22.12 |




## Yield by target population

| Method / cohort | Done | Success / attempts | Success bounds | Unordered relations | Relations / attempt |
|---|---|---|---|---|---|
| buchberger / planted | 24/25 | 24/25 | 96.0–100.0% | 45 | 1.800 |
| buchberger / sampled | 29/30 | 0/30 | 0.0–3.3% | 0 | 0.000 |
| f5b / planted | 22/25 | 22/25 | 88.0–100.0% | 41 | 1.640 |
| f5b / sampled | 29/30 | 0/30 | 0.0–3.3% | 0 | 0.000 |

Success bounds allow all unfinished targets to succeed for the upper endpoint. Relation counts and relations per attempt are observed lower bounds when timeouts exist. No finite upper bound on the unknown relation counts is claimed in this table.

Buchberger recovered 89 ordered / 45 unordered signed relations; f5b recovered 81 / 41. All completed cases matched the independently enumerated golden roots and signed point decompositions, with zero solver errors. No completed sampled target yielded a relation at these small factor-base sizes; the unfinished sampled target in each run remains unknown.

The method comparison describes this fixed fixture set, implementation, machine, and deadline. It does not establish a general ranking of Gröbner algorithms. The sampled cohort also includes five empty-base diagnostics; basis construction and model variant remain separate strata in the JSON rather than being interpreted as an unbiased cryptanalytic population.

## Separate 14-fixture field-coverage batch

| Method | Done / 14 | Solve timeouts | Unordered relations | Σ solved (s) |
|---|---|---|---|---|
| buchberger | 14 | 0 | 10 | 0.804 |
| f5b | 14 | 0 | 10 | 0.828 |

Each method recovered all seven planted targets; all seven sampled targets completed without a relation. Both coverage batches had zero errors or diagnostic timeouts. These are separate runs of the added fixtures, preserving the original 55-case timings above.

# Isogenies and two-term maps

Explicit known answers are distinguished from class-level metadata

| Map battery | Known answers | Scope |
|---|---|---|
| Separable degree-2 isogeny | 20 | E: $y^2=x^3-x$ to E′: $y^2=x^3+4x$; kernel {O,(0,0)} |
| Gaussian unit | 40 | $\varphi(x,y)=(-x, i\,y)$, where i lies in the base field |
| Eisenstein unit | 78 | $\varphi(x,y)=(\zeta_3 x,y)$, where $\zeta_3$ lies in the base field |
| Relative binary Frobenius | 70 | $(x,y) \mapsto (x^2,y^2)$, with the coefficient-squared target model |

The explicit 2-isogeny has $x'=x-1/x$ and $y'=y(1+1/x^2)$ away from its kernel. The kernel and infinity map to infinity; exceptional inputs are tested. A class identified by equal trace can lack an explicit map in this release. The dataset records this distinction and does not supply a complete isogeny graph or volcano.

## Selected-subgroup endomorphism tests

The classifier supports 30 of the 56 core curves, including 6 binary Koblitz Frobenius cases. All 480 scalar decompositions verify a kernel-lattice determinant, the exact congruence $k \equiv k_1+k_2\lambda \pmod r$, nearest-plane bounds, and the signed point reconstruction $[k]G=[k_1]G+[k_2]\varphi(G)$. Unsupported cases carry explicit reasons.

For binary Koblitz coefficients in $F_2$, $\tau(x,y)=(x^2,y^2)$ is an endomorphism of the same curve, with $\tau^2-(2a-1)\tau+[2]=0$. For binary coefficients outside $F_2$, this squaring map ordinarily lands on a different coefficient-squared curve. It must not be used as a same-curve GLV map.

A map must preserve the chosen cyclic subgroup before a subgroup eigenvalue or scalar decomposition is valid. The rank-two examples test this directly: $[r]P=O$ alone cannot establish $P \in \square G \square$. Powers of an endomorphism with a quadratic minimal polynomial do not by themselves supply a balanced four-term GLV/GLS construction.

## Reference timing samples

| Curve | Scalar baseline (µs) | Full two-term path (µs) |
|---|---|---|
| p61-E3-5 | 247.0 | 191.4 |
| p64-E4-2 | 174.9 | 161.8 |
| K83-a0 | 6305.1 | 4338.8 |

Medians of three variable-time Python samples, one operation per sample. The full path includes decomposition, the endomorphism, and joint signed multiplication; precomputed joint-multiplication timings are separately named in JSON. These samples illustrate the timing interface and are separate from polynomial relation-search performance.

# A genuine GLV + GLS4 example

A separate quadratic-extension supplement; the 56-curve corpus is preserved

| Parameter | Verified value |
|---|---|
| Base prime / field | $p=2^{41}-853=2199023254699$; $\mathbb{F}_{p^2}=\mathbb{F}_p[u]/(u^2-2)$ |
| Field cardinality | $q=4835703274706983025580601$ (82 bits) |
| Curve | $E: y^2=x^3+(7+5u)$; ordinary, $j=0$ |
| Rational group | $C_2 \times C_{2r}$; $N=4r$ |
| Prime subgroup | $r=120892581867617635107885 7$ (80 bits) |
| Geometric End ring | $\mathbb{Z}[\zeta_3]$, maximal quadratic order of discriminant $-3$ |

Let $\eta=1+u$. The commuting efficient maps are $\varphi(x,y)=(\zeta_3 x,y)$ and $\psi(x,y)=(\eta^{1-p}x^p, \eta^{3(1-p)/2}y^p)$. The subgroup eigenvalues are $\lambda_\varphi=174560118547837207691337$ and $\lambda_\psi=896867993705181372003749$. On geometric endomorphisms $\psi^2=-\pi q$; on $\mathbb{F}_{p^2}$-rational points $\pi q$ acts as identity, so $\psi^2$ acts as $[-1]$. These are different statements.

The four-term congruence uses eigenweights $1$, $\lambda_\varphi$, $\lambda_\psi$, $\lambda_\varphi\lambda_\psi$. The supplied reduced rank-four lattice has determinant $-r$. Exact rational Gram–Schmidt and Babai nearest-plane decomposition establish universal absolute component ceilings:

```
[754746, 745967, 754745, 745966]
```

Every component is therefore at most 20 bits in absolute magnitude. The accompanying two-term GLV lattice guarantees 40-bit components. Sixteen scalar vectors verify both congruences and actual point reconstructions; the supplement also records field/map answers, certified prime factors, and 51 independent PARI checks.

Four-dimensional refers to a four-term congruence lattice using these GLV and GLS maps. The geometric endomorphism algebra remains quadratic. This supplemental curve has its own scalar/map benchmark family; it is not one of the current 55 polynomial fixtures.

## What remains before a full index-calculus attack benchmark

This release measures two-summand relation solving on finite factor bases. A full ECDLP pipeline additionally needs larger-arity construction, generation-inclusive timing, relation-matrix rank, linear algebra modulo $r$, and individual-log recovery. An instrumented solver must provide maximum working degree before a solving-degree comparison is possible. The Sage/Singular adapter exports actual equations, but no unavailable external-backend timing is fabricated.

References: I. Semaev, Summation polynomials and the discrete logarithm problem on elliptic curves, ePrint 2004/031, eprint.iacr.org/2004/031. P. Longa and F. Sica, Four-dimensional Gallant–Lambert–Vanstone scalar multiplication, ePrint 2011/608, eprint.iacr.org/2011/608. Detailed contracts: docs/ALGEBRA_METRICS.md, docs/BACKEND_PROTOCOL.md, and the canonical JSON artifacts in this release.

# Open leads: what the follow-up found

The original 56 curves, 69 polynomial fixtures, historical measurements, and golden release remain unchanged. This follow-up investigates unresolved solver timeouts, hidden linear constraints, random-target yield, and a concrete isogeny path.

| Lead | New evidence | Status |
|---|---|---|
| GF(2) preprocessing | 57 exact transformations, two-way row-span proofs, and exhaustive original-domain checks | Verified |
| Difficult relation solves | Paired reruns: 24/24 transformed and 21/24 original runs completed at a 20 s budget | Measured |
| Regularity as a grouping feature | Equal affine ideals can have different highest-input-part regularities after row reduction | Generator dependent |
| Sampled-target zero yield | Exact pair-sum supports for 27 factor bases explain the extremely low uniform-target density | Quantified |
| Same order, different groups | An explicit dual pair of degree-two isogenies connects two existing p61 models | Constructive |

The practical improvement is ordinary Gaussian elimination applied before Gröbner computation. Every output equation has an exact combination witness, and every original equation is checked to lie in the output row span. These checks establish equality of polynomial ideals, rather than only agreement on sampled roots.

The timing experiment covers four selected difficult binary cases and two actual SymPy algorithms, with three paired repetitions. It does not establish a universal backend ranking. The original solver’s maximum working degree remains unavailable; no final-basis degree is substituted for it.

# A certified preprocessing step

Collect the monomials occurring in the exact recorded GF(2) equations. Interpret each polynomial as a bit row over those columns, row-reduce with XOR, retain nonzero rows, and put any discovered constant unit first. No monomial multiples or guessed relations are introduced.

For original generators F and reduced generators R, the certificate proves both inclusions of their constant-coefficient row spans:
R ⊆ span(F), and F ⊆ span(R).
Consequently ⟨F⟩ = ⟨R⟩ as affine polynomial ideals. Variables, coordinate domains, factor bases, signed targets, and expected point decompositions are unchanged.

| Case | Input rows | Output rows | Linear output rows | Unit |
|---|---|---|---|---|
| K83-a0-f3-low_monomials-d6-planted | 97 | 29 | 4 | False |
| S83-a1-f3-low_monomials-d6-planted | 97 | 31 | 6 | False |
| S83-a1-f3-low_monomials-d6-seeded_subgroup | 97 | 32 | 6 | True |
| B83-a0-f3-subgroup_seeded_basis-d5-planted | 95 | 31 | 5 | False |

All 57 binary fixtures admit this exact transformation. Their derived systems pass exhaustive evaluation over 35,760 declared assignments, followed by independent subgroup and signed-point checks. Prime-characteristic cases are explicitly rejected by this GF(2)-specific implementation.

Proof masks identify the original equations XORed to obtain each output equation. The verifier also checks reverse inclusion: checking only that outputs are valid combinations would permit dropping necessary equations. Tests reject forged units, false combination witnesses, omitted rows, changed targets, and an unsupported coefficient characteristic.

# Paired solver measurements

Each row below summarizes three new paired repetitions. Both variants use fresh isolated workers, identical targets and finite domains, grevlex, and a 20-second worker budget. Diagnostic regularity work is disabled during these performance pairs. Variant order alternates across pairs.

| Case / method | Complete: raw / RREF | Median raw wall (s) | Median RREF wall (s) | Paired wall speedup |
|---|---|---|---|---|
| K83-a0-f3-low_monomials-d6-planted buchberger | 3/3 · 3/3 | 0.604 | 0.438 | 1.38× |
| K83-a0-f3-low_monomials-d6-planted f5b | 0/3 · 3/3 | — | 3.628 | — |
| S83-a1-f3-low_monomials-d6-planted buchberger | 3/3 · 3/3 | 4.925 | 0.507 | 9.86× |
| S83-a1-f3-low_monomials-d6-planted f5b | 3/3 · 3/3 | 5.272 | 0.651 | 8.10× |
| S83-a1-f3-low_monomials-d6-seeded_subgroup buchberger | 3/3 · 3/3 | 4.402 | 0.479 | 9.18× |
| S83-a1-f3-low_monomials-d6-seeded_subgroup f5b | 3/3 · 3/3 | 4.271 | 0.462 | 9.38× |
| B83-a0-f3-subgroup_seeded_basis-d5-planted buchberger | 3/3 · 3/3 | 1.318 | 0.311 | 4.26× |
| B83-a0-f3-subgroup_seeded_basis-d5-planted f5b | 3/3 · 3/3 | 1.379 | 0.361 | 3.82× |

Wall time starts before preprocessing and exact proof verification and ends after worker completion. It includes interpreter/SymPy startup, serialization, root extraction, and signed relation checks. The paired core metric separately adds preprocessing and proof verification to the worker’s core solve timer. Neither includes factor-base construction or fixture generation.

A dash in a median or speedup means no matched completed observations. Timeouts remain censored; the 20-second budget is not treated as an exact solve duration. Completed-only medians are shown with completion counts. Three repetitions provide local measurements, not confidence intervals or production implementation results.

# Regularity and relation-yield power

| Equivalent affine system | Original top-part regularity | RREF top-part regularity |
|---|---|---|
| K83-a0-f3-low_monomials-d6-planted | 6 | 6 |
| S83-a1-f3-low_monomials-d6-planted | 7 | 6 |
| S83-a1-f3-low_monomials-d6-seeded_subgroup | 7 | 0 |
| B83-a0-f3-subgroup_seeded_basis-d5-planted | 6 | 5 |

Regularity here is the first zero Hilbert-function value of $J = \langle \mathrm{top}(f) : f \text{ is a recorded input generator} \rangle$. Row-equivalent affine generator lists can define different $J$. Therefore this number is meaningful only with a fixed generator list and preprocessing policy; it is not an intrinsic curve invariant.

| Field | Highest-density recorded factor base | Points | Exact-support probability ≈ |
|---|---|---|---|
| b53 | K53-a1 | 4 | 6.63e-07 |
| b79 | K79-a1 | 4 | 8.39e-21 |
| b83 | B83-a0 | 4 | 2.32e-10 |
| p61 | p61-G-0 | 8 | 4.73e-07 |
| p64 | p64-G-0 | 8 | 5.13e-11 |
| p83 | p83-S-0 | 32 | 6.62e-24 |
| p83s | p83s-G-0 | 8 | 1.51e-15 |

For each pool, enumerate every unordered signed pair, combine repeated factors correctly, and exclude identity targets. If $S$ distinct nonzero sums occur in an $r$-order subgroup, an ideal uniform nonzero target has success probability $S/(r-1)$. Across the 37 seeded-case contexts, the uniform-model expected successes total only 1.80005e-06; the same sum bounds the chance of any success.

These are exact support calculations under a hypothetical uniform-target model. The existing hash-seeded targets are deterministic and modulo-biased, so the figure is not a p-value for those recorded trials. The zero-yield observations have very little power to compare curve families. Exact fractions and every supported target are retained in the JSON.

# An explicit isogeny between represented curves

```
Over p = 2^61 − 1, the two p61 E4 records have the same trace and order 2^61, yet their rational groups
differ:
p61-E4-0: Z/2 × Z/2^60
p61-E4-1: Z/2^61.
```

The previously recorded degree-two quotient can be connected all the way to the second stored representative. This closes the gap between a trace-based class-membership statement and a concrete map for this pair.

```
E: y² = x³ − x
E′: Y² = X³ + 4X
φ(x,y) = (x − 1/x, y(1 + 1/x²)).
Its kernel is {O,(0,0)}.
```

The stored target is $y^2 = x^3 - 37x$. Choose $u = 416766181166040365$ modulo $p$, with $4u^4 = -37$. Scale the quotient output by $(u^2,u^3)$ to reach that exact model.

```
For the dual, undo the scale to recover (X,Y), then apply
ψ(X,Y) = ((X + 4/X)/4, Y(1 − 4/X²)/8).
At infinity or X=0, the appropriate map returns infinity.
```

Symbolic rational-function checks verify both curve equations and both dual compositions, $\psi\varphi=[2]$ and $\varphi\psi=[2]$, after the appropriate scaling. The dataset adds 26 explicit point/dual-composition vectors and 340 homomorphism checks, including exceptional points.

This example illustrates the distinction between isogeny and rational group isomorphism. It supplies one verified path between existing records; it does not construct every path or a complete isogeny volcano. The primary curve and class counts are unchanged.

# Recovered work and remaining leads

The timeout follow-up retried 17 historical incomplete solves or diagnostics at a 20-second budget. It recovered 5 of 6 prior solve timeouts; 12 diagnostic runs finished. Every completed relation solve matched the golden roots and signed decompositions. Historical 2.5-second observations remain unchanged.

The consolidated metric catalog now contains exact highest-input-part Hilbert diagnostics and verified final-basis metrics for all 69 original systems. Every value points to an actual completed computation. An algorithm-independent regularity value can be shared across methods; the timing of a censored F5b diagnostic is never filled from a Buchberger run.

| Priority | Open lead | Concrete next experiment |
|---|---|---|
| 1 | Positive random-target controls | Use small-field controls with enough pair-sum density to observe successes; hold subgroup size, factor-base policy, and sampling model fixed. |
| 2 | Actual working-degree instrumentation | Add a precisely defined intermediate-degree trace or instrumented Macaulay/F4 backend; retain unavailable values until measured. |
| 3 | Larger polynomial systems | Extend relation arity and vary subspace dimensions after profiling certified preprocessing; record new case identities and rank/yield. |
| 4 | Independent external backend | Execute the existing Sage/Singular adapter on supported coefficient fields; compare exact roots and actual stage timings. |
| 5 | Isogeny / endomorphism-aware factor bases | Compare matched factor-base pools and targets under explicit maps, including kernel and image restrictions. |
| 6 | Complete discrete-log pipeline | Collect independent relation rows, measure rank and linear algebra modulo r, and verify individual-log recovery before claiming an attack runtime. |

Reproduce from the project checkout:

```
python tools/follow_up_leads.py --phase all --timeout 20 --repetitions 3
python -m unittest discover -s tests -v
python tools/build_followup_report.py
```

Sources and method context: the bundled SymPy 1.14.0 runner; Semaev, ePrint 2004/031; the original report’s Tate/isogeny references. New observations are supported by the exact transformations, pair-sum census, symbolic map checks, and raw measurements in followups/2026-10-15/.